ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ-4 **SUMMER SEMESTER, 2024-2025**

DURATION: 35 MINUTES **FULL MARKS: 15**

Math 4641: Numerical Methods

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

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1. State the fundamental theorems of calculus. Applying these theorems, explain how Euler's method is used to approximate the value of a definite integral. $3+2$ (CO4) (PO1)

Solution:

The fundamental theorems of calculus are —

i. Let f be a continuous real-valued function defined on a closed interval $[a, b]$. Let F be the function defined, for all x in $[a, b]$ by

$$F(x) = \int_a^x f(t)dt$$

Then F is uniformly continuous on $[a, b]$ and differentiable on the open interval (a, b), and $F'(x) = f(x)$ for all x in (a, b) so F is an antiderivative of f.

ii. If f is Riemann integrable on $[a, b]$ then

$$\int_a^b f(x)dx = F(b) - F(a)$$

The idea is to rewrite the integral as the solution of an ODE,

$$\frac{dy}{dx} = f(x), y(a) = 0, \text{ using the 1st Fundamental theorem}$$

where then $y(b)$ gives us the value of the integral $\int_a^b f(x)dx$, according to the 2nd Fundamental theorem. We can use Euler's method to approximate the value of the dependent variable y at the discrete point $x = b$.

Rubric:

- 1.5 points each correct fundamental theorem.
- 2 points for the correct explanation.

2. Compare and contrast the following numerical algorithms for solving 1st order ODEs — Euler's method, Runge-Kutta 2nd order method, and Runge-Kutta 4th order method. 2 (CO3) (PO1)

Solution:

Any 2 valid comparisons will suffice. Some comparisons can be—

Criteria	Euler	Runge-Kutta 2nd Order	Runge-Kutta 4th Order
Formula	$y_{i+1} = y_i + f(x_i, y_i)h$	$y_{i+1} = y_i + (a_1k_1 + a_2k_2)h$, $k_1 = f(x_i, y_i)$, $k_2 = f(x_i + ph, y_i + qk_1h)$	$y_{i+1} = y_i + (a_1k_1 + a_2k_2 + a_3k_3 + a_4k_4)h$
Terms of the Taylor series	First 2 terms.	First 3 terms.	First 5 terms.
Order of the Taylor polynomial	1st order — which is why it's also called the Runge-Kutta 1st order method.	2nd order — which is why it's called the Runge-Kutta 2nd order method.	4th order — which is why it's also called the Runge-Kutta 4th order method.
Local Truncation Error	$O(h^2)$	$O(h^3)$	$O(h^5)$
Global Truncation Error	$O(h)$	$O(h^2)$	$O(h^4)$
Variant Approaches	None.	Has 3 different versions – Heun's method, Midpoint method, and Ralston's method	Has 2 different versions – Runge's approach and Kutta's approach.

Rubric:

- 1 point for each valid comparison.

3. Observe the lightly shaded solid object portrayed in Figure 1.

   Its base is enclosed by a curve $y = e^x - 1$, the y-axis, and the straight line $y = 3$. When this solid object is cut by a plane perpendicular to the y-axis and the xy-plane, the resulting cross-section is a square (as indicated by the darkly shaded cross-sectional area).

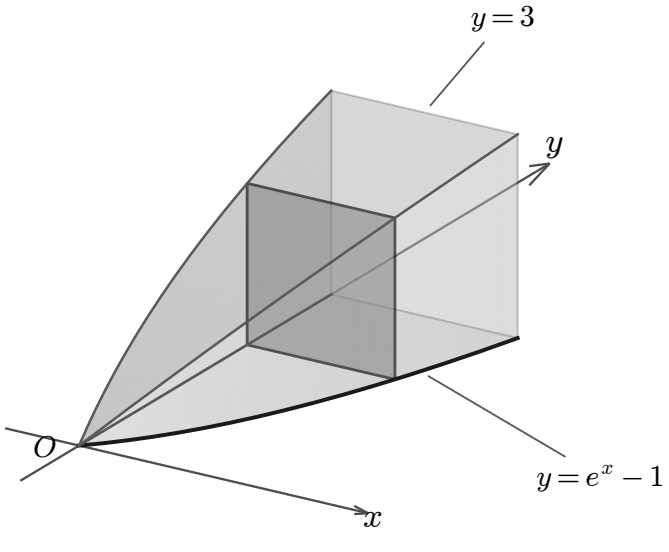


Figure 1: A solid geometric object.

a) Formulate an integrand function $g(y)$ whose integral value within the limits $[0, 3]$ will yield the volume of the solid object. 1.5 (CO1) (PO1)

Solution:

There are 3 key observations here—

- The slicing planes are perpendicular to the y-axis, so the slices are stacked along the y-axis. The variable of accumulation is therefore y, not x.
- The cross-sections of the object are squares.
- The side of each square is measured *horizontally*, from the y-axis upto the curve. So the side length is the x-coordinate of the curve at that particular height y.

We know, that the area of a square having a side of length s is s^2.
Here, the side length must be expressed as a function of y, which requires us to make x the subject of the equation of the curve,

$$y = e^x - 1 \Rightarrow e^x = y + 1 \Rightarrow x = \ln(y + 1)$$

So, for a particular value of y, the side length of an infinitesimally thin square cross-sectional slice is $\ln(y+1)$, and the area of that square slice will be $[\ln(y + 1)]^2$. All of these infinitesimally thin square slices within the limits $[0, 3]$ together constitute the entire volume of the solid object, *ergo* the integrand is,

$$g(y) = [\ln(y + 1)]^2$$

Rubric:

- 1.5 points for a correct integrand function.

b) Approximate the volume of the solid object using Simpson's 1/3 rule, with $n = 6$ segments. 4.5 (CO2) (PO1)

Solution:

Here, $h = \dfrac{b-a}{n} = \dfrac{3-0}{6} = 0.5$. Let's compute the functional values at the right-hand side of each of these $n = 6$ segments.

$$\begin{aligned}
y_0 &= a = 0; & g(y_0) &= 0 \\
y_1 &= a + h = 0 + 0.5; & g(y_1) &= 0.1644 \\
y_2 &= y_1 + h = 1; & g(y_2) &= 0.4805 \\
y_3 &= y_2 + h = 1.5; & g(y_3) &= 0.8396 \\
y_4 &= y_3 + h = 2; & g(y_4) &= 1.2069 \\
y_5 &= y_4 + h = 2.5; & g(y_5) &= 1.5694 \\
y_6 &= y_5 + h = 3; & g(y_6) &= 1.9218
\end{aligned}$$

The multiple-segment equation for Simpson's 1/3 rule is

$$I = \int_a^b g(y)dy \approx \frac{b-a}{3n}\left[g(y_0) + 4\sum_{\substack{i=1,\\ i=\text{odd}}}^{n-1} g(y_i) + 2\sum_{\substack{i=2,\\ i=\text{even}}}^{n-2} g(y_i) + g(y_n)\right]$$

where, $a = 0$ and $b = y_6 = 3$. Plugging in the respective values, we get,

$$\begin{aligned} I &\approx \frac{3}{3 \times 6}\left[0 + 4 \times (0.1644 + 0.8396 + 1.5694) + 2 \times (0.4805 + 1.2069) + 1.9218\right] \\ &= \frac{3}{18}\left[10.2936 + 3.3748 + 1.9218\right] \\ &= 2.5984 \end{aligned}$$

So the volume of the solid object is $I = \int_0^3 [\ln(y+1)]^2\, dy \approx 2.5984$.

Rubric:

- 2 points for the correct functional values at all the nodes.
- 2 points for a correct application of Simpson's 1/3 rule.
- 0.5 point for an approximately correct answer.

4. An internet meme is defined as a humorous or culturally relevant image, video, or text that spreads rapidly online, often evolving as it is shared and adapted by users. Draw a rough sketch of a math meme related to this course. 2 (CO69) (PO420)
Warning: Cringe and lame memes will result in negative marking.

Solution:



Figure 2: Click on the thumbnail image.

Rubric:

- 2 points for a dank meme.